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Perimeter & Area

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: A rectangle is 10 cm long and 7 cm wide. What is its area and perimeter?
Method & answer: Area = 10 × 7 = 70 cm². Perimeter = 2×(10+7) = 34 cm.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A rectangle is 7 cm by 5 cm. What is its perimeter?

2. A square has sides of 8 cm. What is its perimeter?

3. A rectangle is 12 cm by 4 cm. What is its area?

4. Which shape has a perimeter of 28cm? A) a 10cm×4cm rectangle B) a square of side 7cm C) both A and B D) neither

5. A rectangle is 15 cm long and 3 cm wide. What is its perimeter?

6. A square has a perimeter of 24 cm. What is the length of one side?

7. What is the area of a 6cm by 6cm square? A) 12cm² B) 24cm² C) 36cm² D) 48cm²

8. A square has an area of 81 cm². What is the length of one side?

9. Select ALL of these shapes that have a perimeter of exactly 24cm: A) an 8cm×4cm rectangle B) a 9cm×3cm rectangle C) a 7cm×5cm rectangle D) a square of side 6cm E) a 10cm×2cm rectangle

10. A rectangle has a length of 11 cm and a width of 4 cm. What is its perimeter?

11. A rectangle is 13 cm long and 2 cm wide. What is its area?

12. Which rectangle has the larger area? A) 8cm×3cm B) 5cm×5cm C) they are equal D) cannot tell

13. A square has sides of 10 cm. What is its perimeter?

14. A rectangle is 7 cm by 7 cm. What shape is it, and what is its area?

15. A rectangle has an area of 48 cm² and a width of 6 cm. What is its length?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A triangle has a base of 8 cm and a height of 5 cm. What is its area?

2. Which of these has an area of 24cm²? A) a triangle with base 8cm, height 6cm B) a triangle with base 12cm, height 4cm C) both A and B D) neither

3. A triangle has a base of 9 cm and a height of 8 cm. What is its area?

6. What is the area of a triangle with base 14cm and height 4cm? A) 18cm² B) 28cm² C) 56cm² D) 36cm²

7. A rectangular garden is 8 m long and 7 m wide. What is its area?

8. Select ALL of these triangles that have an area of MORE than 20cm²: A) base 9cm, height 5cm B) base 6cm, height 7cm C) base 8cm,

11. Two rectangles, 5cm×4cm and 3cm×4cm, are joined with no overlap. What is the total area?

12. A square lawn has a perimeter of 44 m. What is its area?

13. A rectangle's length is three times its width. If the width is 3 cm, what is the perimeter?

height 4cm D) base 10cm, height 5cm E) base 5cm, height 8cm

4. An L-shape is a 8cm×5cm rectangle with a 3cm×2cm square cut from one corner. What is its area?

5. A rectangular field is 35 m by 20 m. What is its perimeter?

9. A triangle has an area of 27 cm² and a base of 9 cm. What is its height?

10. A rectangle is 1.5 m long and 60 cm wide. What is its area in cm²?

14. A rectangle has a perimeter of 38 cm and a length of 11 cm. What is its area?

15. An L-shaped garden is a 12m×7m rectangle with a 4m×3m rectangle removed from a corner. What is its area?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A rectangular field is 30 m by 14 m. A path 1 m wide is built all the way around the inside edge. What is the area of the path?

2. A rectangular lawn is 16 m by 10 m. A 5m×3m flower bed is cut from one corner. What is the remaining area of the lawn?

3. Which shape has the LARGEST area? A) a triangle base 12cm height 9cm B) a 7cm×8cm rectangle C) a 6cm×9cm rectangle D) a square of side 7cm

4. A triangle and a rectangle have equal areas. The rectangle is 10 cm by 5 cm, and the triangle has a base of 10 cm. What is the triangle's height?

5. A rectangular room measures 7 m by 5 m. Carpet costs £13 per m². What is the total cost of carpeting the room?

6. A square garden has an area of 169 m². What is the perimeter of the garden?

7. Select ALL of these statements that are TRUE about a triangle with base 16cm and height 5cm: A) its area is 80cm² B) its area is 40cm² C) it is exactly half of a 16cm×5cm rectangle D) its area cannot be found without more information

8. A 28cm×18cm piece of card has a 3cm×3cm square cut from each of its four corners. What is the remaining area?

9. A rectangle's length is 6 cm more than its width. If its perimeter is 36 cm, what are its dimensions?

10. A large rectangle 13m×9m has a 5m×3m rectangle removed from a corner, forming an L-shape. What is the perimeter of the L-shape?

11. A rectangular sports pitch is 70 m long and 45 m wide. A running track goes around the outside with a uniform width of 2 m. What is the perimeter of the outer edge of the track?

12. A square and a rectangle have equal areas. The square has sides of 9 cm and the rectangle's width is 3 cm. What is the rectangle's length?

13. An L-shaped room is made of two rectangles, 6m×4m and 3m×2m (no overlap). Carpet costs £14 per m². What is the total cost to carpet the room?

14. A rectangular banner's length and width are both doubled. If the original area was 24 cm², what is the new area?

15. A triangle has a base of 18 cm and a height that is a third of the base. What is its area?